TileSmith

LED Wall Pixel Map & Test Pattern Generator · User Manual

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Pixel mapping, LED engineering, NDI®, Spout and Syphon output and an animated technician who builds the wall, tile by tile.

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1. What TileSmith does

TileSmith lays out LED walls on a master canvas, fills them with test patterns and pixel maps, works out processor ports, data paths and power, and sends the canvas live over **NDI®** to your media server or processor. An animated LED technician can build each wall tile by tile, which you can show live or record as a video.

- Several LED walls on one canvas, each with its own tile model, size, position and test pattern.
- Processor-aware engineering: tiles per port and data paths (P# to R#), plus tiles per circuit and power runs on Socapex.
- Live NDI® output, plus Spout (Windows) or Syphon (Mac) to other apps on the same computer (desktop app); HAP, MP4 and ProRes video export, PNG, CSV, spec sheet and Resolume Advanced Output XML.
- Runs fully offline. Everything you set up is saved automatically.

> The desktop app (Windows or Mac) is needed for NDI®, Spout and Syphon output and video export. The web version does everything else.

2. Installing and first launch

Build instructions come with the source as **BUILD-WINDOWS.txt** and **BUILD-MAC.txt**. In short:

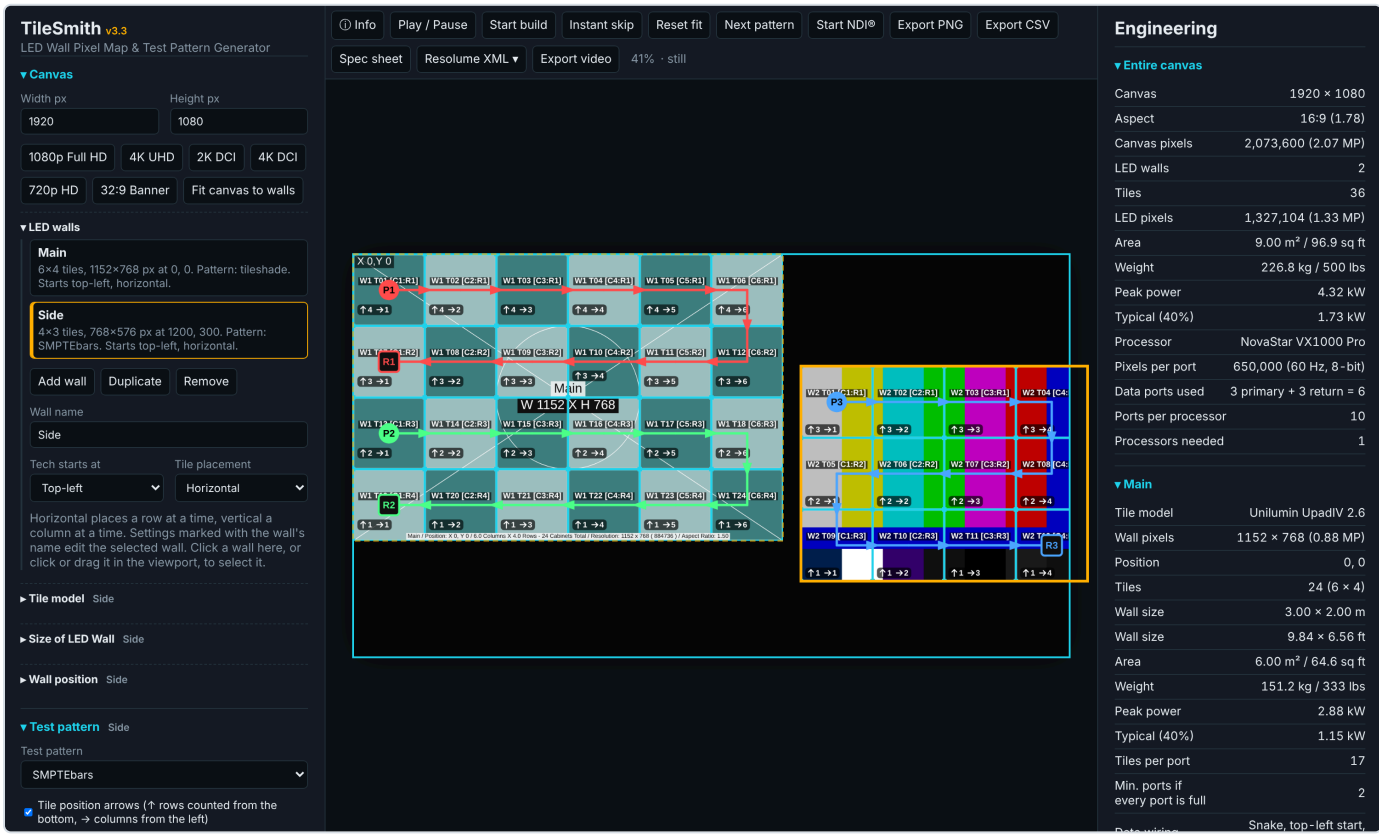
1. On the build computer, install **Node.js** (LTS) and the free **NDI® SDK** (ndi.video/for-developers/ndi-sdk). The build copies the NDI® runtime and its license notices from the SDK into TileSmith, so computers that *run* TileSmith need nothing NDI® installed.
2. For video export, add ffmpeg: on Windows, the gyan.dev "full" build's `ffmpeg.exe` goes in the project's `bin` folder; on Mac, install it with Homebrew.
3. In the project folder run `npm run clean`, `npm install`, then `npm run dist` (Windows) or `npm run dist:mac` (Mac). Spout (Windows) and Syphon (Mac) come with the project in its `bin` folder and are packed in automatically; there is nothing extra to download or install.
4. Windows gives you **TileSmith-Portable.exe**, a single file you can copy anywhere. Mac gives you a .dmg containing a universal app.
5. The Android and iOS apps are built from the project's `mobile` folder with Android Studio or Xcode: see **BUILD-ANDROID.txt** and **BUILD-IOS.txt**, and chapter 21.

The window title and the header show the version (**v3.3**), so you can confirm you are running the latest build.

3. Quick start

1. Canvas: pick the canvas size your media server outputs, for example 1080p.
2. Size of LED Wall and Tile model: set columns and rows and choose your tile preset.

3. Test pattern: choose a pattern, or click **Next pattern** in the top bar to step through them. **tileshade** is a classic pixel map.
4. Engineering › LED processor: choose your processor to see tiles per port, then pick a pattern under Data wiring.
5. Engineering › Power: tick **Power distribution** to see tiles per circuit, then pick a pattern under Power wiring.
6. Click **Start NDI®** in the top bar (desktop app) and pick *TileSmith-LED-Out* in your receiver. For an app on the same computer, click **Start Spout** (Windows) or **Start Syphon** (Mac) and pick *TileSmith*.
7. Click **Start build** to watch the technician build the wall, or **Instant skip** to show it finished.

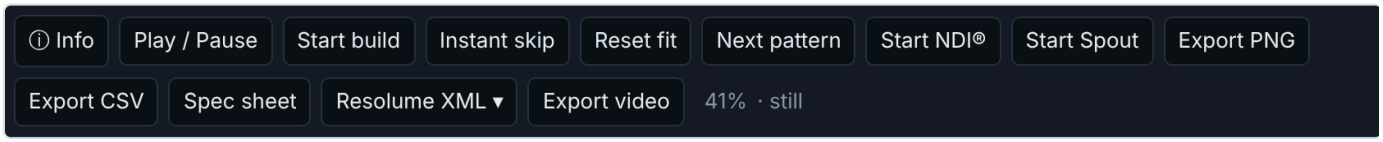


The TileSmith window: settings on the left, actions along the top, the master canvas in the middle and the engineering panel on the right.

4. The screen

Left panel holds every setting, in collapsible sections, in this order: Canvas (with LED walls, Tile model, Size of LED Wall and Wall position submenus), Test pattern (with a Tile borders & overlays submenu), Engineering (with LED processor and Power submenus; Data wiring is a submenu inside LED processor and Power wiring a submenu inside Power), Animation Options (with DVD screensaver, Build options, Video export and NDI® options submenus, plus Spout options on Windows or Syphon options on Mac; Technician is a submenu inside Build options), and Label printing options (below Animation Options; it appears only once data or power wiring is defined). Click a heading or submenu to open or close it. Data wiring appears once a processor is chosen; the power options and Power wiring appear once **Power distribution** is ticked.

Sections marked with a wall name edit the selected wall. **Top bar** holds the actions. **Viewport** in the middle shows the master canvas exactly as it is sent over NDI®. **Engineering** on the right shows totals for the whole canvas and for each wall.



Top bar of the Windows app. The Mac app shows Start Syphon in place of Start Spout; the web version has neither.

Button	What it does
ⓘ Info	Credits, contact, Ko-fi link, the NDI®, Spout and Syphon notices and **Reset saved settings**.
Play / Pause	Pauses or resumes the build animation.
Start build	Clears the walls and starts the technicians building from the first tile.
Instant skip	Shows every wall fully built.
Reset fit	Fits the whole canvas in the viewport again.
Next pattern	Moves the selected wall to the next test pattern in the list, and back to the first after the last. *custom* is skipped until you upload an image.
Start NDI®	Starts sending the canvas as an NDI® source (desktop app). While it runs the button is green and reads **Stop NDI®**; click it again to stop. Settings are in Animation Options › NDI® options.
Start Spout / Start Syphon	**Start Spout** on Windows, **Start Syphon** on Mac: shares the canvas with other apps on the same computer, straight from the graphics card. Green and **Stop ...** while live. Shown only in a desktop app built with Spout or Syphon. Settings are in Animation Options › Spout options or Syphon options.
Export PNG	Saves the canvas as a PNG image at full resolution.
Export CSV	Saves a patching sheet with one row per tile.
Spec sheet	Opens a printable spec sheet with every engineering figure.
Resolume XML ▾	Drop-down: pick **One slice per LED wall** or **One slice per tile** to save an Advanced Output preset for Resolume Arena.
Export video	Records the build to a video file (desktop app). Click again to stop and discard.

Moving around the viewport

- Next to the zoom level, the top bar shows how many frames per second the canvas is drawing, or *still* when nothing is moving.
- **Scroll wheel** zooms around the pointer. Zoom all the way out and the whole canvas, and any wall placed outside it, is centred in view, however small the window. **Drag empty space** pans. **Reset fit** brings the canvas back into view.
- In a small window the top bar stays on one row; scroll it sideways to reach the rest of the buttons.
- **Click a wall** to select it. **Drag a wall** to move it (see LED walls).

5. Canvas

The **Canvas** section also holds the **LED walls**, **Tile model**, **Size of LED Wall** and **Wall position** submenus described in the next chapters.

The master canvas is the full raster TileSmith outputs over NDI® and in exports. Type a **Width** and **Height**, or click a preset: 1080p Full HD (1920 × 1080), 4K UHD (3840 × 2160), 2K DCI (2048 × 1080), 4K DCI (4096 × 2160), 720p HD (1280 × 720), 32:9 Banner (3840 × 1080).

Fit canvas to walls shrinks the canvas to the area your walls use, so no empty pixels are sent. This also lowers CPU use for NDI®.

The cyan outline in the viewport is the edge of the canvas, and each wall has an amber outline (solid on the selected wall when there are several). Anything outside the walls is sent as black.

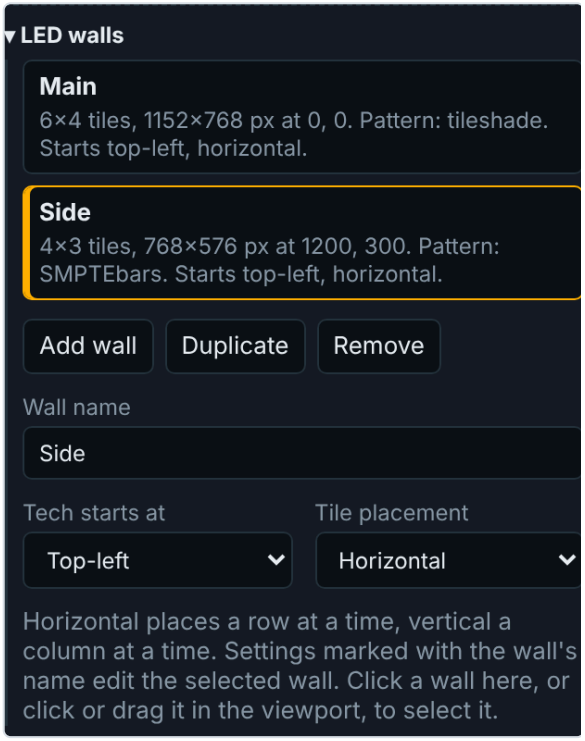


6. LED walls

The **LED walls** list shows each wall with its size, position, pattern and build start. Click one to select it. **Add wall** places a new wall beside the others, **Duplicate** copies the selected wall, **Remove** deletes it.

Wall name appears in the list, the engineering panel, exports and on the tileshade pattern.

Tech starts at picks the corner the technician starts from. **Tile placement** builds a row at a time (horizontal) or a column at a time (vertical).

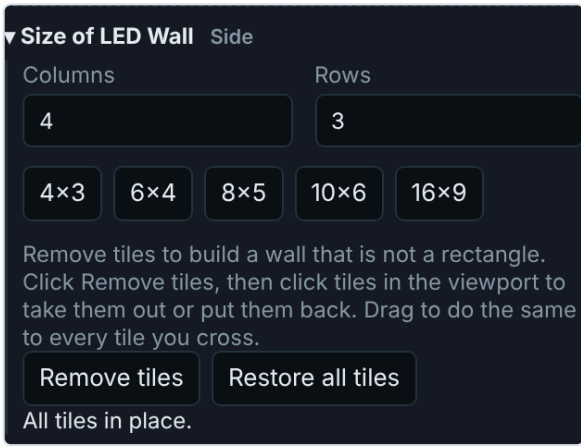


Positioning walls

Under **Wall position**, type **Offset X** and **Offset Y** in canvas pixels, or use the alignment buttons. In the viewport you can **drag a wall**: it snaps to the canvas edges and to other walls, keeping a **2 px gap** between walls. Hold **Alt** while dragging to place it freely. Pink guide lines show what it is snapping to.

Size of LED Wall

Set **Columns** and **Rows** of tiles (up to 64 each), or click a quick size such as 6×4 or 16×9. The wall's pixel size is columns × tile width by rows × tile height.



Removing tiles

For a wall that is not a rectangle, click **Remove tiles** in the Size of LED Wall submenu, then click tiles in the viewport to take them out. Click a removed tile to put it back, or press and drag to do the same to every tile you cross. **Restore all tiles** puts every tile back. While you are removing tiles, the viewport marks removed tiles with a red × (this never appears in the output).

A removed tile is black in the output and left out everywhere else: the build animation skips it, and the engineering panel, spec sheet, CSV and Resolume per-tile slices leave it out. Tile IDs keep their grid numbers, so the remaining tiles keep their names. Data and power wiring skip removed tiles, and Snake and Linear join neighbouring ports or circuits that now fit on one. Removing tiles works with any model. On a wall with half tiles, Remove tiles takes out whole tiles, and clicking an empty space puts a tile back (a full tile if its whole cell is empty, otherwise a half tile).



A ROE Carbon CB5 wall with the top corners stepped in.

7. Tile model

Presets are grouped by manufacturer. Pick a **Preset** to fill in the tile's pixels, size, pitch, depth, weight and maximum watts, or edit any field (the preset then shows *Custom*). **Density** and **Typical (40%)** power are worked out for you. These figures drive the engineering panel, port capacity and power circuits.

ROE Carbon CB5 Half and CB8 Half are estimated from the full panel (ROE publishes only the CB3 Half): half the pixels and watts, and the CB3 Half's weight. YES TECH figures come from the manufacturer's spec sheets, which give power per square metre: **Max W** is the higher figure times the cabinet area, rounded up. Check them against your exact model and brightness version before planning power.



Preset	Tile (px)	Size (mm)	Pitch	Weight	Max W
ROE Carbon CB3	160 × 320	600 × 1200	3.75 mm	13.2 kg	540
ROE Carbon CB3 Half	160 × 160	600 × 600	3.75 mm	7.5 kg	260
ROE Carbon CB5	104 × 208	600 × 1200	5.77 mm	13.85 kg	500
ROE Carbon CB5 Half	104 × 104	600 × 600	5.77 mm	7.5 kg	250
ROE Carbon CB8	72 × 144	600 × 1200	8.33 mm	12.68 kg	430
ROE Carbon CB8 Half	72 × 72	600 × 600	8.33 mm	7 kg	215
ROE Ruby RB2.3	216 × 216	500 × 500	2.31 mm	8.16 kg	180
ROE Black Pearl BP2V2	176 × 176	500 × 500	2.84 mm	9.35 kg	190
Absen Polaris PL2.5 Pro	200 × 200	500 × 500	2.5 mm	9.6 kg	128
Unilumin UpadIV 2.6	192 × 192	500 × 500	2.6 mm	6.3 kg	120
INFiLED Deep Black DB2.6	192 × 192	500 × 500	2.6 mm	8 kg	170
YES TECH MG10 1.9	256 × 256	500 × 500	1.95 mm	6.6 kg	113
YES TECH MG10 1.9 (1000-1200 nit)	256 × 256	500 × 500	1.95 mm	6.6 kg	140
YES TECH MG10 2.6	192 × 192	500 × 500	2.6 mm	6.6 kg	113
YES TECH MG10 2.6 (1000-1200 nit)	192 × 192	500 × 500	2.6 mm	6.6 kg	140
YES TECH MG9 2.9	168 × 168	500 × 500	2.98 mm	6.3 kg	175

Preset	Tile (px)	Size (mm)	Pitch	Weight	Max W
YES TECH MG9 3.9	128 × 128	500 × 500	3.91 mm	6 kg	175
YES TECH MU 1.9	256 × 256	500 × 500	1.95 mm	7.6 kg	163
YES TECH MU 2.6	192 × 192	500 × 500	2.6 mm	7.6 kg	113
YES TECH MU 2.6 500×1000	192 × 384	500 × 1000	2.6 mm	12.6 kg	225
YES TECH MU 3.9	128 × 128	500 × 500	3.91 mm	7.6 kg	105
YES TECH MU 3.9 500×1000	128 × 256	500 × 1000	3.91 mm	12.6 kg	210
YES TECH MU 4.8	104 × 104	500 × 500	4.81 mm	7.6 kg	105
YES TECH MU 4.8 500×1000	104 × 208	500 × 1000	4.81 mm	12.6 kg	210
YES TECH MU 2.9 outdoor	168 × 168	500 × 500	2.98 mm	7.6 kg	175
YES TECH MU 2.9 outdoor 500×1000	168 × 336	500 × 1000	2.98 mm	12.6 kg	350
YES TECH MU 3.9 outdoor	128 × 128	500 × 500	3.91 mm	7.6 kg	175
YES TECH MU 3.9 outdoor 500×1000	128 × 256	500 × 1000	3.91 mm	12.6 kg	350
YES TECH Xtra 1.9	256 × 256	500 × 500	1.95 mm	7.5 kg	163
YES TECH Xtra 2.6	192 × 192	500 × 500	2.6 mm	7.5 kg	163
YES TECH Xtra 2.9 outdoor	168 × 168	500 × 500	2.98 mm	7.5 kg	175
YES TECH Xtra 3.9 outdoor	128 × 128	500 × 500	3.91 mm	7.5 kg	175
YES TECH MG6S Plus 1.9	256 × 384	500 × 750	1.95 mm	8.3 kg	150
YES TECH MG6S Plus 2.6	192 × 288	500 × 750	2.6 mm	8.3 kg	169
YES TECH MG6S Plus 2.8	176 × 264	500 × 750	2.84 mm	8.2 kg	158
YES TECH MG6S Plus 3.9	128 × 192	500 × 750	3.91 mm	7.6 kg	158
YES TECH MG6S Plus 5.2	96 × 144	500 × 750	5.21 mm	7.5 kg	158
YES TECH MG6S 2.6	192 × 192	500 × 500	2.6 mm	6.7 kg	150
YES TECH MG6S 2.84	176 × 176	500 × 500	2.84 mm	6.2 kg	120
YES TECH MG6S 3.9	128 × 128	500 × 500	3.91 mm	5.8 kg	130
YES TECH Mega 3.9 outdoor	128 × 256	500 × 1000	3.91 mm	12.5 kg	400
YES TECH Mega 4.6 outdoor	108 × 216	500 × 1000	4.63 mm	12.5 kg	400
YES TECH Mega 5.7 outdoor	88 × 176	500 × 1000	5.68 mm	12.5 kg	400
YES TECH MAir 5.9 mesh outdoor	168 × 168	1000 × 1000	5.95 mm	19 kg	500
YES TECH MAir 7.8 mesh outdoor	128 × 128	1000 × 1000	7.81 mm	19 kg	500
YES TECH MAir 5.9 solid outdoor	168 × 168	1000 × 1000	5.95 mm	23.5 kg	500
YES TECH MT II 3.9-7.8 1000×500	256 × 64	1000 × 500	3.91 mm	8.7 kg	225

Preset	Tile (px)	Size (mm)	Pitch	Weight	Max W
YES TECH MT II 3.9-7.8 500×500	128 × 64	500 × 500	3.91 mm	5.8 kg	113

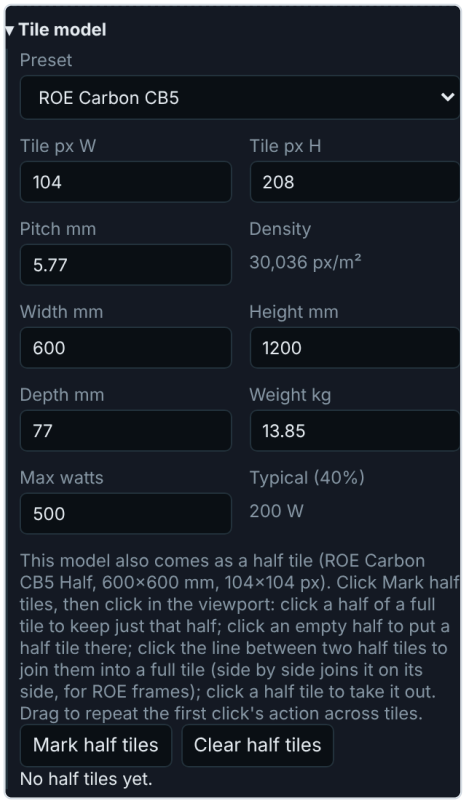
Half tiles

Some models also come as a half tile, so a wall can step in or out, or shift, by half a tile. When the wall's model has a half version, the Tile model submenu shows **Mark half tiles**. Click it, then click in the viewport:

- Click one half of a full tile to keep just that half; the other half becomes an empty space.
- Click an empty half space to put a half tile there.
- Click the line between two half tiles to join them into one full tile. Halves above each other make an upright tile, which can also sit across two rows, half a tile lower than its neighbours. For ROE Carbon panels, two halves side by side join into a full panel on its side, to match ROE's frame options.
- Click a half tile to take it out.

Press and drag to repeat the first click's action across tiles. While marking, empty half spaces show a red × in the viewport (never in the output). **Clear half tiles** goes back to a plain grid of full tiles. The button is hidden for models without a half version and for the half models themselves; pick a half model as the preset to build a whole wall from half tiles.

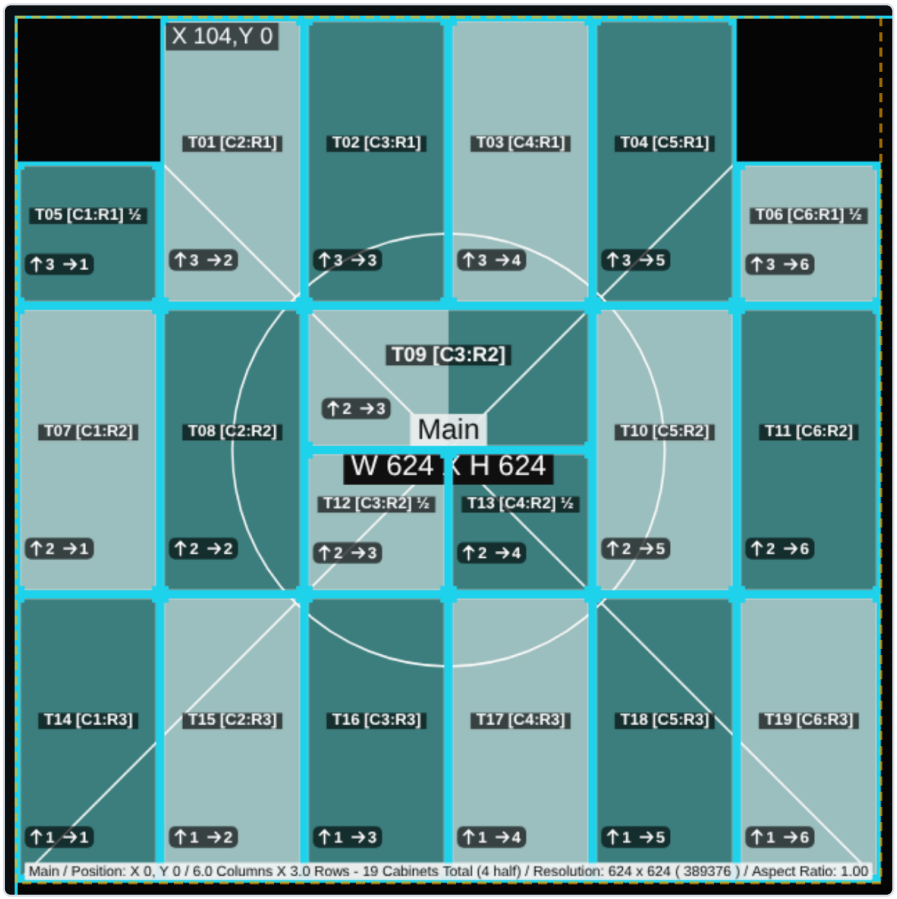
Model	Half tile	Half
ROE Carbon CB3	ROE Carbon CB3 Half (600×600 mm, 160×160 px)	top / bottom
ROE Carbon CB5	ROE Carbon CB5 Half (600×600 mm, 104×104 px)	top / bottom
ROE Carbon CB8	ROE Carbon CB8 Half (600×600 mm, 72×72 px)	top / bottom
YES TECH MU 2.6 500×1000	YES TECH MU 2.6 (500×500 mm, 192×192 px)	top / bottom
YES TECH MU 3.9 500×1000	YES TECH MU 3.9 (500×500 mm, 128×128 px)	top / bottom
YES TECH MU 4.8 500×1000	YES TECH MU 4.8 (500×500 mm, 104×104 px)	top / bottom
YES TECH MU 2.9 outdoor 500×1000	YES TECH MU 2.9 outdoor (500×500 mm, 168×168 px)	top / bottom
YES TECH MU 3.9 outdoor 500×1000	YES TECH MU 3.9 outdoor (500×500 mm, 128×128 px)	top / bottom
YES TECH MT II 3.9-7.8 1000×500	YES TECH MT II 3.9-7.8 500×500 (500×500 mm, 128×64 px)	left / right



Each tile is drawn where it really sits: a half tile fills half a cell, a full tile on its side spans two columns. Tile IDs are

numbered across the top row first and half tiles end in ½; the build animation carries each tile at its own size; and the engineering panel, spec sheet, CSV (column `Tile`) and Resolume per-tile slices use each tile's own size, pixels, weight and watts. The layout is kept if you switch to a model without a half version and comes back when you switch back.

Ports and circuits count every tile by its own pixels and watts and show their load in pixels or watts. On a wall with half tiles, Snake and Linear follow the pattern across the half-tile grid and start a new port or circuit wherever the next tile would overload it; the Data wiring preference and the power Wiring preference do not apply.



A ROE Carbon CB5 wall: the top row steps in by half a tile at each end, and the middle has a panel on its side over two half panels.

8. Test patterns

Each wall has its own pattern. Choose it from **Test pattern**, or click **Next pattern** in the top bar to step the selected wall through the list in this order. **Tile-shade color** appears right under the pattern list when tileshade or tileshade-clean is chosen. **Tile position arrows** (an option for every pattern, on by default)

labels every tile with its row counted up from the bottom (↑) and its column counted from the left (→), so ↑1 →1 is the bottom-left tile. The marker sits low on the left of each tile and moves up or down on any tile where it would cover the tile labels, the wall's name and size boxes or the footer. Only lit tiles show the pattern, so during a build it appears tile by tile.

Pattern	What it shows
tileshade	Pixel map. Every tile is a light or dark shade of one colour (set it with **Tile-shade color**), with corner-to-corner diagonals, a centre circle, the wall name and resolution in the middle, the canvas position of the first tile's top-left corner in a see-through dark box, and a footer with the wall name, position, columns, rows, cabinet count, resolution and aspect ratio. This is the default pattern.
tileshade-clean	The same pixel map without the name, size and position boxes on the wall; that information is only in the footer.
tileshade-cycle	The tileshade layout with every tile stepping through red, green and blue. Each colour holds for a second, then the next one slides in from the left over 1.5 seconds. Each tile starts at a different point in the sequence, so a tile never shows the same colour as the tiles above, below, left or right of it. Good for spotting dead pixels, colour mismatches and swapped tiles. Tile-shade color is not used.
SMPTEbars	Classic SMPTE colour bars, drawn by TileSmith at 1920 × 1080. Image fit and Scroll apply.
ramp	16-step greyscale from black to white, left to right.
sweep	Moving rainbow with a light-to-dark vertical fade (animated).
white	Full-field white.
red	Full-field red.
green	Full-field green.
blue	Full-field blue.
black	Full-field black.
gray50	Full-field 50% grey.
smptescroll	SMPTE bars image. Picking it switches Scroll to diagonal if Scroll was off.
disguise test pattern 16:9	Built-in 1920 × 1080 image with grid, circles, greyscale steps and colour references.
disguise test pattern 4:3	Built-in 1024 × 768 version of the same image.
branded test pattern 16:9	Built-in 1920 × 1080 colour bars with a black centre box. Put your own text or logo in the box with **Center box**.
branded test pattern 4:3	Built-in 1024 × 768 version with the same centre box.

Pattern	What it shows
custom	Your own image: click **Upload test pattern**. Images keep their full size up to 8192 px. Image fit and Scroll apply.
pixel1	Single-pixel checkerboard (every other pixel on). Good for checking pixel mapping and receiving-card setup.
circles	Circles and crosshairs for checking geometry and aspect ratio.
gradH	Smooth black-to-white gradient, left to right.
gradV	Smooth black-to-white gradient, top to bottom.
rgbramp	Four ramps from black to red, green, blue and white.
zone	Zone plate (concentric rings) for spotting scaling and moiré.
cycle	Full-field colour that changes every 1.5 seconds (animated).

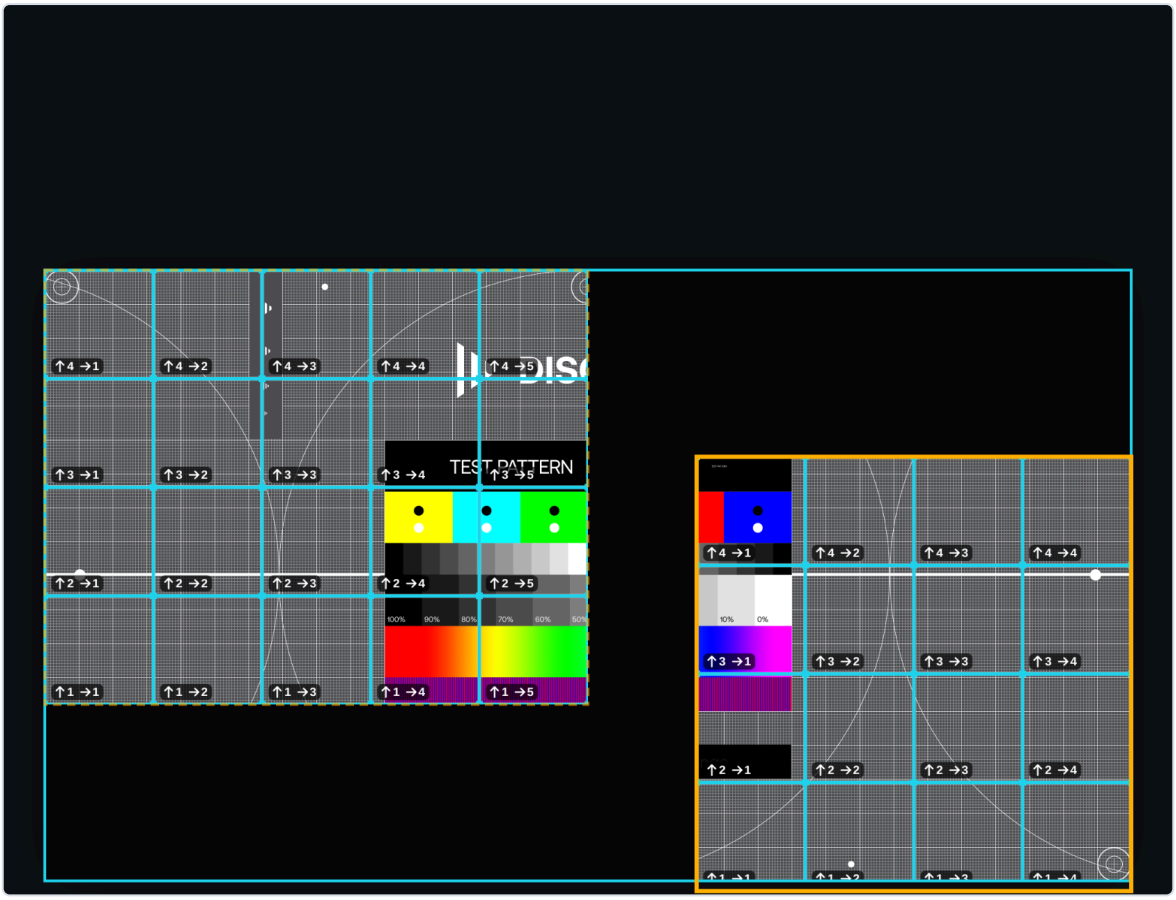
Scroll

Scroll moves any pattern across the wall (none, diagonal ↘, diagonal ↗, horizontal →, vertical ↓); **Scroll speed** sets how fast. Tile borders and labels stay still while the pattern moves underneath.

Image fit

For image patterns (SMPTEbars, smptescroll, the built-in images and your own upload), **Image fit** decides how the image meets the wall:

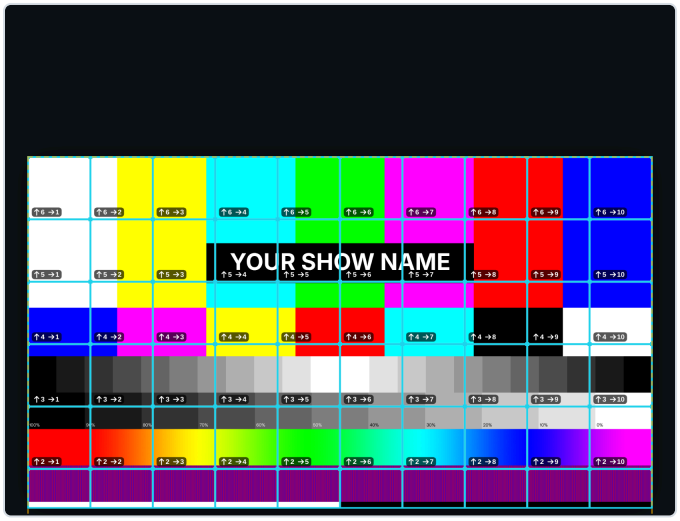
Image fit	What it does
Stretch to wall	Fills the wall exactly. The image is stretched if its shape differs from the wall.
Fit inside wall	Shows the whole image without stretching; black bars fill the gaps.
Fill wall (crops edges)	Fills the wall without stretching; the edges that do not fit are cut off.
Original size on canvas	No scaling. The image sits at the top-left corner of the master canvas and each wall shows the part of the image underneath it, like content coming from a media server. Use this when your image was made at canvas resolution.
Original size on wall	No scaling. The image starts at the wall's own top-left corner.



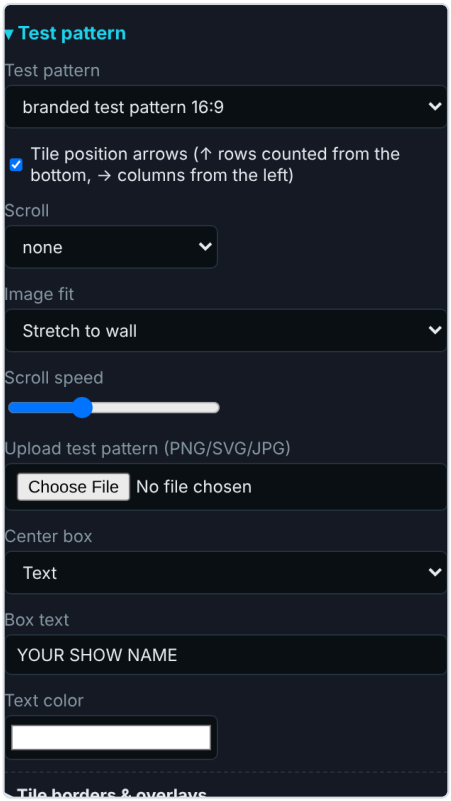
Original size on canvas: two walls each show their own part of one 1920 × 1080 image, with nothing scaled.

Centre box on the branded patterns

With a **branded test pattern** selected, **Center box** puts your **Text** (with a colour) or a **Logo** (upload PNG, SVG or JPG) in the black box. Text shrinks to fit; logos keep their shape, even when the pattern is stretched.



Branded test pattern with text in the centre box.



Test pattern settings.

Tile-shade color sets the colour used by the tileshade pattern.

9. Tile borders and overlays

These settings are a submenu inside **Test pattern**.

Border color and **Thickness** set the outline drawn around each tile, with small corner marks.

Tile IDs labels each tile, for example *T07 [C1:R2]* (with the wall number when there are several walls). **Pixel address ranges** shows each tile's X and Y pixel range. **Millimeter marks** shows the tile's physical size.



10. LED processor

These settings are the **LED processor** submenu inside **Engineering**. Choose the processor (or sending card) that drives your walls. TileSmith then shows how many tiles each port can drive and enables data wiring. With **None**, the data and port options are hidden.

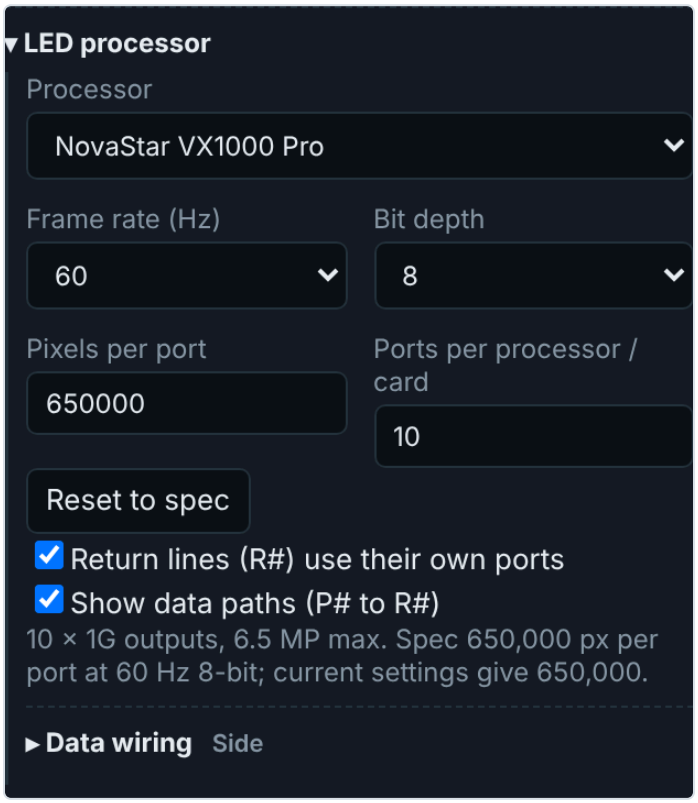
- **Frame rate** and **Bit depth** adjust the pixels per port the way each manufacturer specifies (for example, NovaStar halves capacity at 10-bit; Brompton scales with frame rate and bit depth).

- **Pixels per port** and **Ports per processor / card** can be edited for your exact system; **Reset to spec** restores the published figures.
- **Ultra Low Latency** (Brompton) halves port capacity.
- **Return lines (R#) use their own ports** counts a return port for every primary port.
- **Show data paths** draws the wiring on the canvas.

Processor	Ports	Pixels per port (spec)	Notes
Brompton Tessera S4	4	525,000 at 60 Hz, 8-bit	4 × 1G outputs.
Brompton Tessera S8	8	525,000 at 60 Hz, 8-bit	8 × 1G outputs, 4.5 MP max.
Brompton Tessera SX40 + 4 XD	40	525,000 at 60 Hz, 8-bit	4 × 10G to Tessera XD units, 10 ports each. 4096 × 2160 max.
NovaStar VX400 Pro	4	650,000 at 60 Hz, 8-bit	4 × 1G outputs, 2.6 MP max.
NovaStar VX1000 Pro	10	650,000 at 60 Hz, 8-bit	10 × 1G outputs, 6.5 MP max.
NovaStar VX2000 Pro	20	650,000 at 60 Hz, 8-bit	20 × 1G outputs, 13 MP max.
NovaStar MX40 Pro	20	650,000 at 60 Hz, 8-bit	20 × 1G outputs (40-port mode via 10G optical), 9 MP max.
NovaStar H series: H_20xRJ45 card	20	650,000 at 60 Hz, 8-bit	H series sending card (H2/H5/H9/H15/H20), 2 slots: 20 × 1G outputs, 13 MP per card, max 10,752 px wide or high, up to 144 Hz. 320,000 px per port at 10-bit. The engineering panel shows how many cards the canvas needs.
NovaStar H series: H_16xRJ45+2xFiber card	16	650,000 at 60 Hz, 8-bit	H series sending card, 2 slots: 16 × 1G outputs, 10.4 MP per card, max 10,240 px wide or high, up to 144 Hz. 320,000 px per port at 10-bit. Fiber 1 copies Ethernet ports 1–8 and fiber 2 copies 9–16, so the fibers add no capacity.
NovaStar H series: H_4xFiber card	32	650,000 at 60 Hz, 8-bit	H series sending card: 4 × 10G fiber outputs, each equal to 8 Ethernet ports through a fiber converter (32 ports), 20.8 MP per card, max 16,384 px wide or high, up to 144 Hz. Figures are for independent mode; copy and backup modes give less. 320,000 px per port at 10-bit.
HELIOS Jr	8	425,000 at 60 Hz, 12-bit	8 × 1G outputs. HELIOS spec is 425,000 px per port at 12-bit 60 Hz.

Processor	Ports	Pixels per port (spec)	Notes
HELIOS 8K	72	425,000 at 60 Hz, 12-bit	8 × 10G to distributors, 72 ports, 34 MP max.
ROE Evision eV4	40	655,360 at 60 Hz, 8-bit	4 × 10G to EVX10 distributors, 40 ports, 8.3 MP max.
Custom	16	650,000 at 60 Hz, 8-bit	Enter your processor's pixels per port and port count.

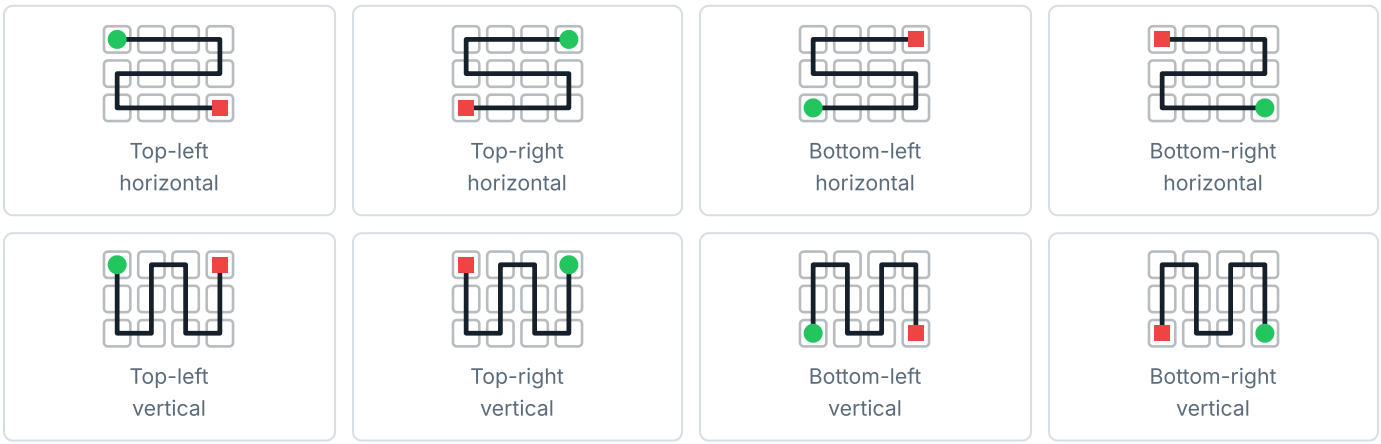
> The engineering panel says how many processors (or sending cards) the canvas needs, using both the port count and the processor's maximum pixel load.



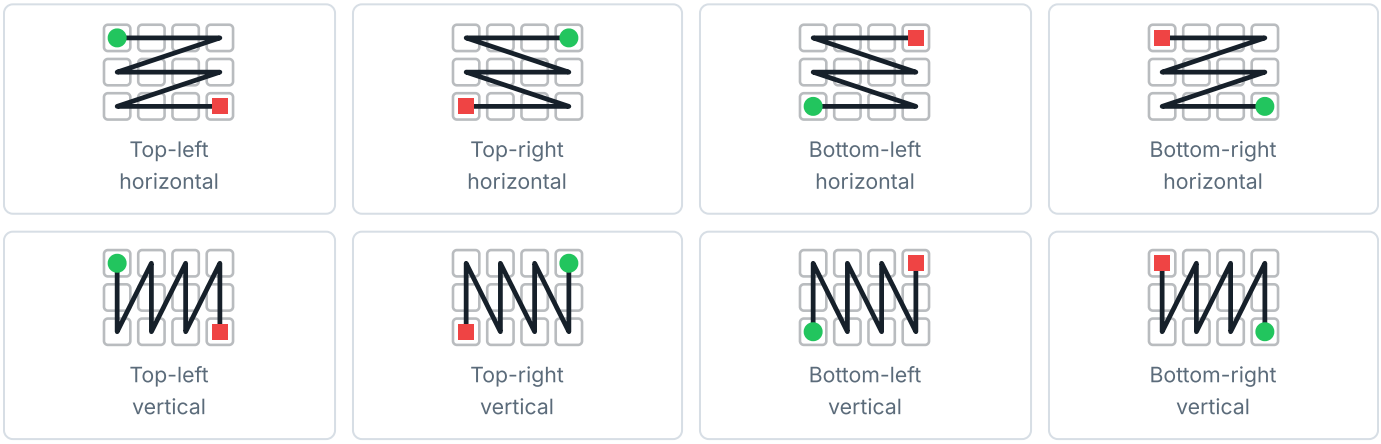
11. Data wiring

Data wiring is a submenu inside **LED processor**. It appears once a processor is chosen and edits the selected wall. Pick a **Pattern**: **Snake** turns back at the end of each line; **Linear** starts every line from the same side; **Manual** lets you click or drag across the tiles in order. Then pick the start corner and direction: the same 8 choices as NovaLCT. In the picker, green is the first tile and red the last.

Snake patterns



Linear patterns



Data wiring preference

Data wiring preference decides how the wall is shared between ports; the pattern then runs inside each port's share. **Horizontal** and **Vertical** keep each port to bands of whole rows or whole columns, whichever direction the pattern runs.

Data wiring preference	How tiles are shared between ports
Fewest ports	Each port gets a rectangle of tiles, sized to use as few ports as possible. The pattern repeats inside every rectangle, like NovaLCT's quick connection.
Whole rows or columns	Each port takes complete rows (complete columns when vertical). If a port cannot drive a whole line, the line is split evenly between ports.
One continuous path	One path runs through the whole wall and is cut wherever a port is full, so a port can start partway along a line.
Horizontal	Each port takes a horizontal band: complete rows across the whole wall, as many as it can drive, whichever way the pattern runs. With a vertical pattern the path snakes up and down inside each band. If a port cannot drive a whole row, each row is split evenly between ports.

Data wiring preference	How tiles are shared between ports
Vertical	Each port takes a vertical band: complete columns down the whole wall, as many as it can drive, whichever way the pattern runs. With a horizontal pattern the path snakes back and forth inside each band. If a port cannot drive a whole column, each column is split evenly between ports.



Snake from the top-left, horizontal, Fewest ports: four ports, each snaking through its own block. P# marks where data enters, R# where it ends.

Data paths are solid lines with arrows and P#/R# markers; power runs are dashed lines marked with a lightning bolt. When a wall shows both, data runs left of each tile's centre and power right of it, in white on a black casing.

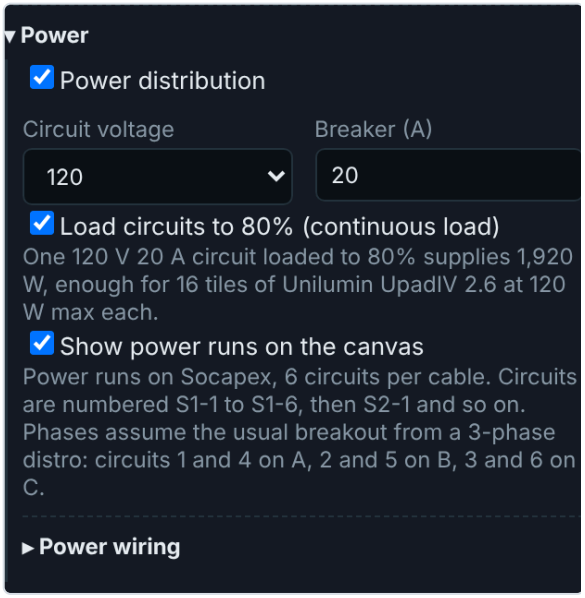
Manual wiring

1. Set Pattern to **Manual** and click **Wire tiles**.
2. Click tiles in the viewport in data order, or press on a tile and drag across the others. A fast drag fills in every tile it crosses; dragging back over the path takes tiles off again. Pressing on the last tile of the path carries on from there. A port stops accepting tiles when it is full.
3. **New port** starts the next port, **Undo** removes the last tile, **Clear** starts again. Click **Stop wiring** when done.

Below the controls TileSmith lists every port (for example *P1→R1: 15/17*), flags ports over capacity in red, and warns about unwired tiles.

12. Power and power wiring

Power works like the LED processor. The settings are the **Power** submenu inside **Engineering**, and nothing about circuits is shown until you tick **Power distribution**. Power is planned on Socapex cables, 6 circuits per cable.



Circuit voltage and **Breaker (A)** set the size of one circuit. With **Load circuits to 80%** on (the usual rule for continuous loads), only 80% of it is used. Tiles per circuit is the usable watts divided by the tile model's **Max watts**, rounded down. For example, 120 V × 20 A × 80% = 1,920 W, which powers 12 tiles of 150 W.

Show power runs on the canvas draws each circuit as a dashed line in its own colour, with a lightning bolt and the circuit number at the first tile and a small square at the last. Circuits are numbered in order across the canvas: 1 to 6 are S1-1 to S1-6, 7 is S2-1, and so on.

When a wall shows both data and power paths, data runs to the left of each tile's centre and power to the right (and a little lower), so the two never sit on top of each other, and power is drawn white on a black casing so it stands out from the coloured data paths (a circuit over capacity stays red). With only one shown, it runs through the tile centres in its usual colours.

Socapex numbering and phases

A Socapex cable carries 6 circuits. TileSmith numbers them *S1-1* to *S1-6* on the first cable, then *S2-1* and so on, and counts the cables needed and the spare circuits left over. Phases assume the usual breakout from a 3-phase distro: circuits 1 and 4 on phase A, 2 and 5 on B, 3 and 6 on C. Breakouts vary, so check this against yours. Circuit numbers continue from one wall to the next.

Power wiring

Power wiring is a submenu inside **Power**. It appears once Power distribution is ticked and edits the selected wall. It works like data wiring: pick **Snake**, **Linear** or **Manual**, then the start corner and direction from the same 8 choices. **Wiring preference** decides how the wall is shared between circuits, with the same choices as the data wiring preference.

Wiring preference	How tiles are shared between circuits
Fewest circuits	Each circuit gets a rectangle of tiles, sized to use as few circuits as possible. The pattern repeats inside every rectangle.

Wiring preference	How tiles are shared between circuits
Whole rows or columns	Each circuit takes complete rows (complete columns when vertical). If a circuit cannot power a whole line, the line is split evenly between circuits.
One continuous path	One path runs through the whole wall and is cut wherever a circuit is full, so a circuit can start partway along a line.
Horizontal	Each circuit takes a horizontal band: complete rows across the whole wall, as many as it can power, whichever way the pattern runs. If a circuit cannot power a whole row, each row is split evenly between circuits.
Vertical	Each circuit takes a vertical band: complete columns down the whole wall, as many as it can power, whichever way the pattern runs. If a circuit cannot power a whole column, each column is split evenly between circuits.

Power link limit caps tiles per circuit at the most tiles your tile model allows on one power daisy chain (see the manufacturer's spec), even when the breaker could take more. 0 means no limit.

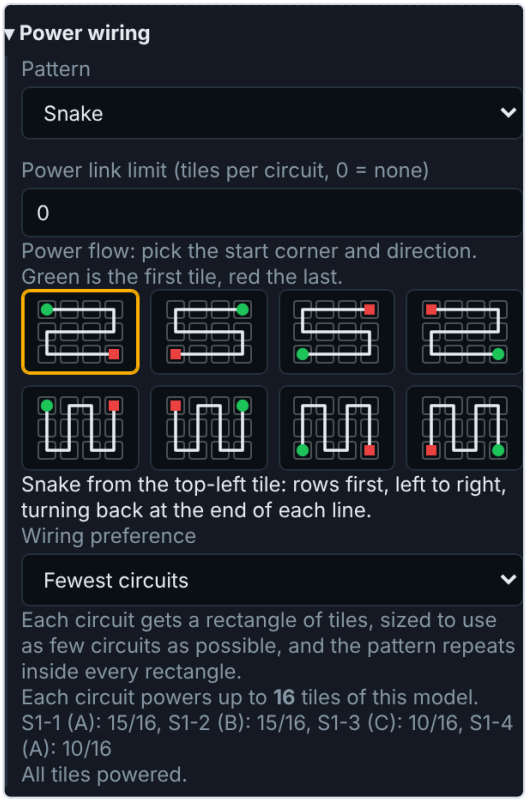


Socapex power runs on a 10 × 5 wall of 150 W tiles at 120 V 20 A. Each circuit powers up to 12 tiles.

Manual power wiring

1. Set Pattern to **Manual** and click **Wire tiles**.
2. Click tiles in the viewport in power-link order, or press on a tile and drag across the others, the same way as data wiring. A circuit stops accepting tiles when it is full.
3. **New circuit** starts the next circuit, **Undo** removes the last tile, **Clear** starts again. Click **Stop wiring** when done.

Below the controls TileSmith lists every circuit with its phase (for example *S1-2 (B): 12/12*), flags circuits over capacity in red and counts tiles without power.

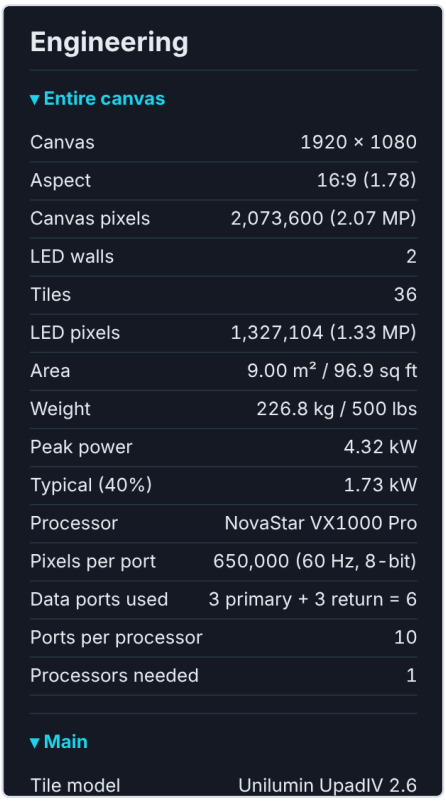


13. Engineering panel

The right-hand panel has an **Entire canvas** section and one section per wall. Click a heading to fold it away.

- Canvas size, aspect, pixels, tile count, LED pixels, area (m² and sq ft), weight (kg and lbs).
- Peak and typical power, and the peak current at the circuit voltage once Power distribution is ticked.
- With Power distribution ticked: circuit size, tiles per circuit, circuits used, the fewest circuits possible, Socapex cables and spare circuits, peak load per phase, circuits over capacity and tiles without power.
- With a processor: pixels per port, tiles per port, ports used (with returns), processors or cards needed, and any ports over capacity.

Spec sheet in the top bar opens the same figures as a printable page.



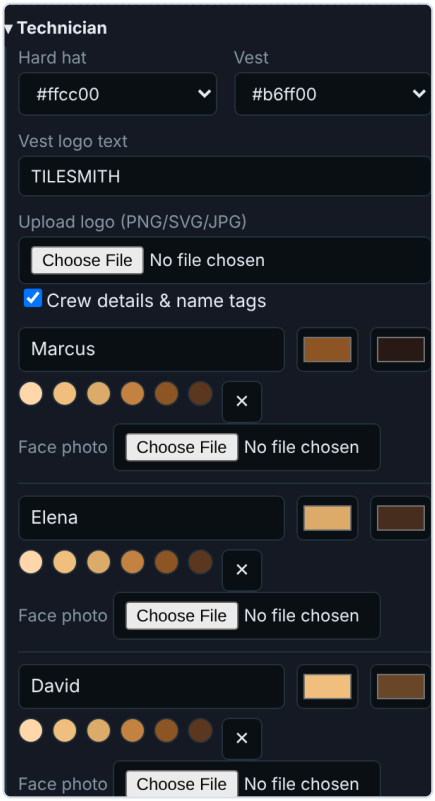
14. Technician and crew

These settings are the **Technician** submenu inside **Animation Options › Build options**.

Choose the **Hard hat** and **Vest** colours, and the **Vest logo text** or an uploaded logo.

Turn on **Crew details & name tags** to name each technician and set skin and hair colour. Add a **Face photo** to put a real face on a technician. **Add crew member** grows the crew; **Reset to defaults** restores it.

With several walls, technicians rotate so the same person is never on every wall at once, and each wall builds at a slightly different rate.



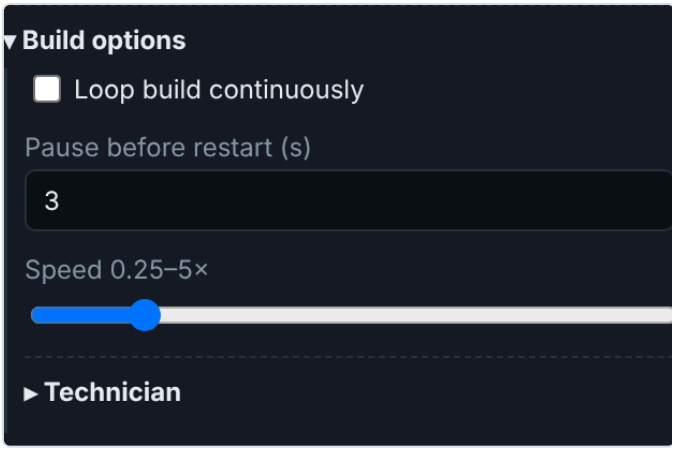
15. Animation options

Start build clears the walls; technicians walk on, carry each tile to its place, latch it and walk off. Only built tiles light up, in NDI® and in exports. Use **Play / Pause** to pause and **Instant skip** to finish.

Under **Animation Options › Build options**: **Loop build continuously** restarts the build after **Pause before restart** seconds; **Speed** runs the build from 0.25× to 5×.



A wall partway through its build.



Build options, with the Technician submenu.

DVD screensaver

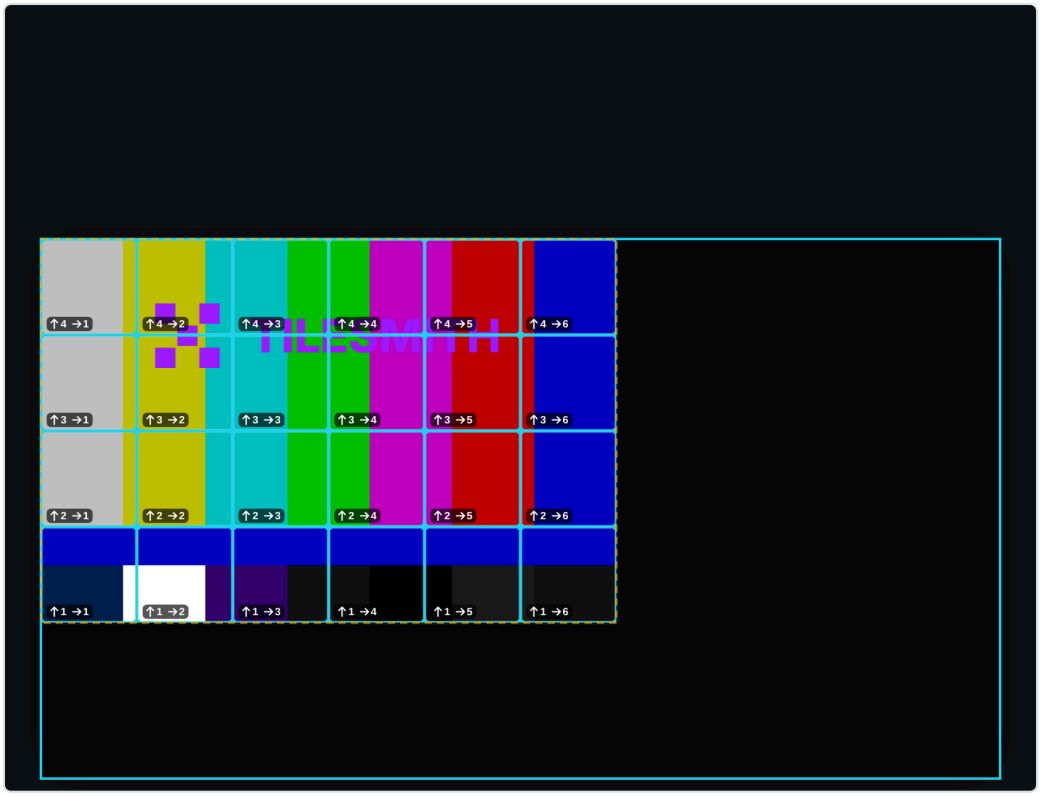
In the **DVD screensaver** submenu, tick **Bounce a logo around the screen** to bounce a logo like the classic DVD screensaver. It shows on every lit tile and is included in NDI® and video export.

- **Logo**: TileSmith logo, Text, Uploaded image. Upload a PNG with a transparent background for the best result.
- **Bounce inside**: **Each LED wall** gives every wall its own logo; **Whole canvas** bounces one logo across the canvas, visible wherever it crosses a wall.
- **Logo size** (percent of the wall's shorter side) and **Bounce speed**.
- **Corner hit chance**: how often the logo lands exactly in a corner. At 100% about 45 of every 100 bounces are corners, at 50% about 22, and at 0% the logo is steered so it never touches a corner. The value shows the expected share of bounces, and the line underneath counts real corner hits as it runs.
- **Change color on each bounce** recolours the logo at every bounce. Turn it off to keep your logo's own colours.

Perfect loop video

Tick **Perfect loop** to make the bounce repeat exactly: after **Loop length** seconds the logo is back where it started, moving the same way and in the same colour, so the video can play on repeat with no visible seam. **Include corner hits** gives two exact corner hits per loop; untick it and the logo never touches a corner. The corner hit chance slider is not used in this mode. TileSmith picks the nearest speed and angle that close the loop, so the speed can differ a little from Bounce speed; the line underneath shows the real speed, bounces and corner hits per loop. A short loop on a very wide wall can force a fast logo, so use a longer loop to slow it down.

Export loop video (desktop app) renders exactly one loop, frame by frame at 60 fps, in the Format chosen under Video export, and saves it next to your other videos as `tilesmith-dvd-loop-…`. The walls are shown fully built, and moving patterns and Scroll are held still so the loop is seamless. Click the button again to cancel.

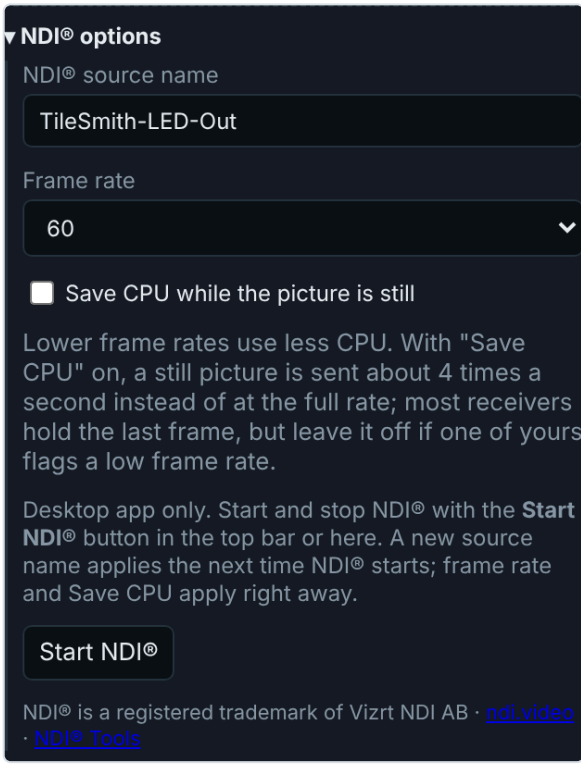


DVD screensaver over SMPTE bars.

16. NDI®, Spout and Syphon output

NDI®

In the desktop app, click **Start NDI®** in the top bar (or in the **NDI® options** submenu inside **Animation Options**). The button turns green and reads **Stop NDI®** while NDI® is live; click it again to stop. The canvas is sent as an NDI® source named after **NDI® source name** (default *TileSmith-LED-Out*). Pick it in Resolume, vMix, OBS, NDI® Studio Monitor or an NDI® decoder. If NDI® cannot start, the reason appears in a message at the bottom of the window.



- **NDI® source name**: change it before you start; a new name applies the next time NDI® starts.
- **Frame rate**: 60, 59.94, 50, 30, 29.97, 25. Lower rates use less CPU. Applies right away.
- **Save CPU while the picture is still**: when nothing moves, sends about 4 frames a second instead of the full rate. Most receivers hold the last frame; leave it off if yours flags a low frame rate.
- The status line in NDI® options shows the source name, the actual frames per second, the resolution and the time each frame takes: **render** (building the frame), **read** (reading the canvas back, including drawing its pixels) and **send** (NDI® compressing and sending it). If the frame rate drops, these show which part is slow.

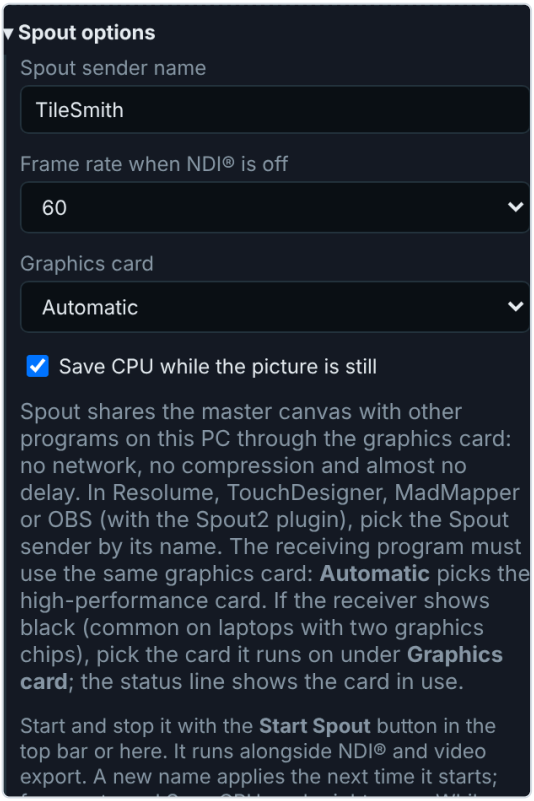
NDI® keeps running at full speed when the window is minimised or behind other windows, and the computer will not nap the app while NDI® output or a recording runs. The NDI® runtime is built into TileSmith, so nothing needs installing: the status line says *built-in NDI® runtime*. A build made without the NDI® SDK falls back to an NDI® Runtime or NDI® Tools installed on the computer and says *installed NDI® Runtime*. If other computers cannot see the source, allow TileSmith through the firewall (Windows: Private networks; Mac: allow Local Network the first time you start NDI®).

Spout (Windows) and Syphon (Mac)

Spout and Syphon share the master canvas with other apps on the **same computer** through the graphics card: no network, no compression and very little CPU. Spout exists only on Windows and Syphon only on Mac, so the Windows app has **Start Spout** and **Spout options**, and the Mac app has **Start Syphon** and **Syphon options**. Use NDI® to reach other computers; NDI®, Spout or Syphon and video export can all run at the same time.

Click **Start Spout** or **Start Syphon** in the top bar (or in its options submenu inside **Animation Options**). The button turns green and reads **Stop Spout** or **Stop Syphon** while it is live; click it again to stop. Then pick *TileSmith* in the receiving app.

- **Spout sender name** / **Syphon server name** (default *TileSmith*): change it before you start; a new name applies the next time output starts. If another Spout sender already uses the name, Spout adds _1, _2 and so on; the status line shows the real name.
- **Frame rate when NDI® is off**: 60, 50, 30, 25. Applies right away. While NDI® runs, Spout and Syphon send at the NDI® frame rate, so every output shows the same frames.
- **Save CPU while the picture is still** (on by default): a still picture is sent about 4 times a second instead of at the full rate. Spout and Syphon receivers keep showing the last frame, so it is safe to leave on.
- **Graphics card** (Spout only): Spout passes the picture through the graphics card, so the receiving app must use the **same card** as TileSmith's sender. **Automatic** (the default) uses the high-performance card, which is where Resolume and most media servers run. If the receiver lists TileSmith but shows black, pick the card the receiver runs on; changing it while Spout runs restarts the sender on that card.
- The status line shows the name, the frames sent each second, the resolution, the time each frame takes to send and, for Spout, the graphics card in use. Syphon says *no app receiving yet* until an app connects; TileSmith only sends Syphon frames while at least one app is receiving. If the canvas size changes, the output follows it.

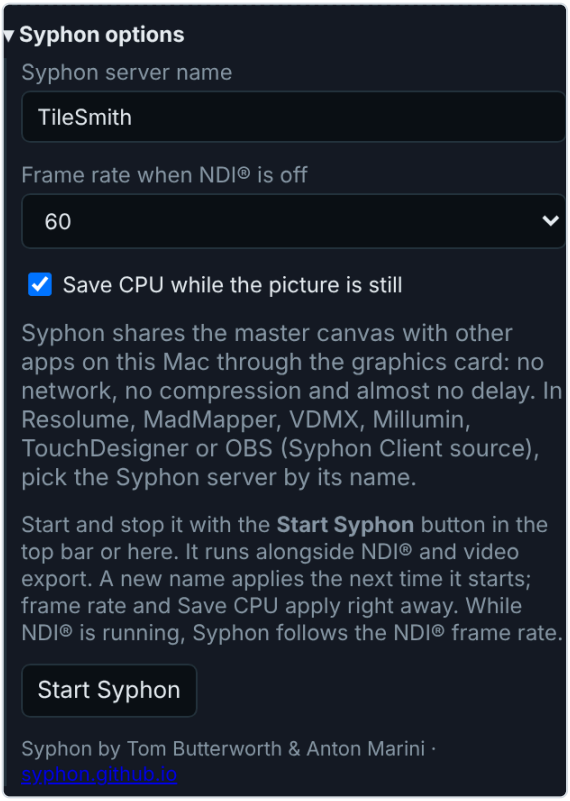


Spout options in the Windows app while Spout is running.

The button and the options appear in a desktop app built with Spout or Syphon. Both come with the project (`bin\spout` and `bin/mac/syphon`) and are built in automatically; a copy built without them hides the button and says so in the options. The web version shows one combined Spout / Syphon options submenu for reference and never has the button. Spout needs nothing installed (no Visual C++ runtime).

Receiving app	Windows (Spout)	Mac (Syphon)
Resolume Arena / Avenue	Sources › Spout › TileSmith	Sources › Syphon › TileSmith
TouchDesigner	Syphon Spout In TOP	Syphon Spout In TOP
OBS Studio	free Spout2 plugin, then a *Spout2 Capture* source	*Syphon Client* source
MadMapper, Millumin, VDMX	MadMapper: Spout input	Syphon input; some list the app as *TileSmith Helper (Renderer)*, which is TileSmith

> On a laptop with two graphics chips (for example Intel and NVIDIA), the receiver must be on the same chip as TileSmith's Spout sender, or it shows black. Leave **Graphics card** on *Automatic* (the fast chip) and make sure the receiver uses the fast chip too (Windows: Settings › System › Display › Graphics, *High performance*), or pick the receiver's chip in the list. Spout needs DirectX 11; if it cannot start, update the graphics driver. Spout keeps a log at `%APPDATA%\TileSmith\spout-log.txt`.



Syphon options in the Mac app.

NDI® licensing and trademarks

NDI® is a registered trademark of Vizrt NDI AB. Learn more about NDI® at https://ndi.video/; the full NDI® Tools (Studio Monitor, Webcam, Screen Capture and more) are at https://ndi.video/tools/.

The desktop app includes the NDI® runtime library from the NDI® SDK (Processing.NDI.Lib.x64.dll on Windows, libndi.dylib on Mac), © Vizrt NDI AB, distributed under the NDI® SDK License Agreement (ndi.link/ndisdk_license). The runtime's third-party license notices ship with TileSmith; open them with ⓘ **Info › View NDI® license notices**. NDI® Tools are not included with TileSmith.

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Spout and Syphon licences

Spout output uses SpoutDX from the Spout SDK by Lynn Jarvis (spout.zeal.co). Syphon output uses the Syphon framework by Tom Butterworth and Anton Marini (syphon.github.io) and node-syphon by Benoît Lahoz. Their licence texts ship inside TileSmith (open them with ⓘ **Info › View Spout / Syphon licences**) and are repeated here.

Spout SDK: SpoutDX (Simplified BSD licence)

Included in the Windows build (SpoutDX.dll, Spout SDK 2.007.017). https://github.com/leadedge/Spout2

SpoutSDK is released under the Simplified BSD licence.

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Syphon Framework (BSD license)

Included in the Mac build (Syphon.framework 5). https://github.com/Syphon/Syphon-Framework

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node-syphon (MIT License)

Included in the Mac build (syphon.node from node-syphon 1.5.5). https://github.com/benoitlahoz/node-syphon

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17. Video export

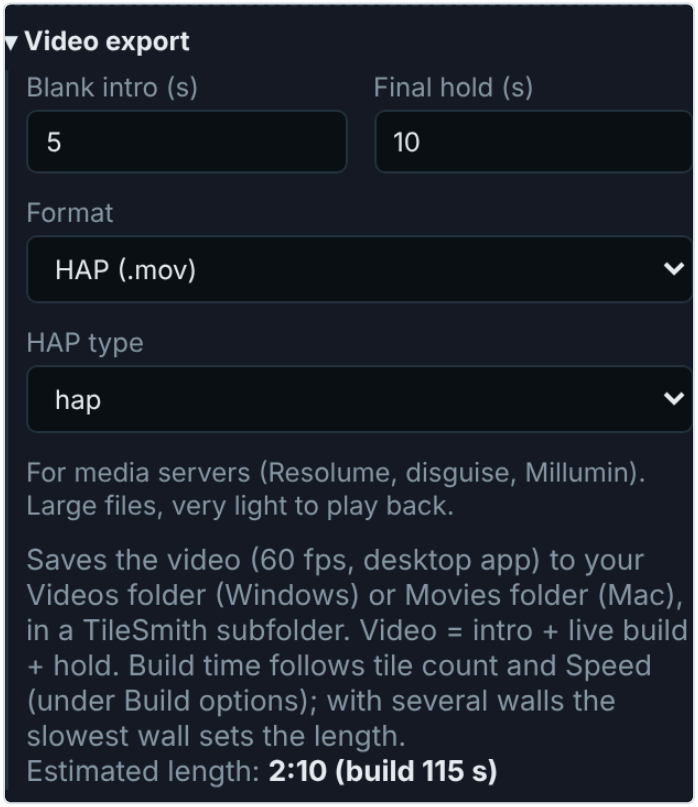
These settings are the **Video export** submenu inside **Animation Options**.

Export video (desktop app) records a 60 fps video: the **Blank intro** (seconds of empty canvas), the whole live build, then the **Final hold** on the finished wall. The length follows the tile count and Speed; with several walls the slowest wall sets it. The estimate is shown under the settings. Files go to `Videos\TileSmith` (Windows) or `Movies/TileSmith` (Mac) and the folder opens when done. Click the button again while recording to cancel.

Format	File	ffmpeg encoder	Best for
HAP (.mov)	.mov	hap	For media servers (Resolume, disguise, Millumin). Large files, very light to play back.
H.264 (.mp4)	.mp4	libx264	Plays almost anywhere and makes small files. Colour detail is shared between pairs of pixels (4:2:0), so single-pixel colour lines soften.

Format	File	ffmpeg encoder	Best for
H.265 / HEVC (.mp4)	.mp4	libx265	Smaller than H.264 at the same quality; needs a newer player. 4:2:0 colour like H.264.
ProRes 422 HQ (.mov)	.mov	prores_ks	Editing master for Premiere, Final Cut, Resolve. Large files, 10-bit 4:2:2.
ProRes 4444 (.mov)	.mov	prores_ks	Highest-quality editing master with full colour on every pixel (4:4:4). Largest files.

For HAP, **HAP type** picks `hap`, `hap_alpha` or `hap_q` (higher quality). All formats are tagged BT.709 so colours match the canvas. Video export needs ffmpeg with the matching encoder (see Installing).



18. Other exports

- **Export PNG**: the full master canvas as an image.
- **Export CSV**: one row per tile with these columns: `Wall`, `TileID`, `Col`, `Row`, `BuildStep`, `StartX`, `StartY`, `EndX`, `EndY`, `W_px`, `H_px`, `Port`, `PortPos`, `Circuit`, `CircuitPos`, `Phase`, `MaxW`, `TypW`, `Tile`. Tile says full, half (with its side) or full on its side; removed tiles have no row. Port and PortPos come from the data wiring; Circuit, CircuitPos and Phase from the power wiring.
- **Spec sheet**: printable engineering summary for the canvas and each wall. In the mobile app it is saved as an HTML file; open it in a browser to print it or save it as a PDF.
- **Print labels**: cable labels for both ends of every data and power cable; see **Cable labels** below.

- **Resolume XML**: an Advanced Output preset whose composition matches the canvas, with one slice per wall or per tile (pick one from the **Resolume XML ▾** drop-down). On the desktop app it saves straight into `Documents/Resolume Arena/Presets/Advanced Output` when that folder exists. In Arena, open Output › Advanced Output, choose the preset, then assign the screen to your real output.

Cable labels

When data wiring or power wiring is defined, a **Label printing options** section appears below Animation Options (it stays hidden until at least one data port or power circuit exists). It prints a label for **both ends of every cable**, so each cable can be found at the processor or distribution box and at the tile, and the two ends of one cable can be matched.

- **Data cables**: the primary cable of each port is labelled **P1, P2 …**; when **Return** (backup) cables are on in LED processor, each port also gets an **R1, R2 …** cable. The *processor end* label names the processor port and the first (or, for R, the last) tile it connects to; the *tile end* label names the tile input (**DATA IN** or **DATA OUT (return)**) and where the cable comes from. Tile numbers are the T01, T02 … numbers on the canvas, with column and row.
- **Power cables**: every circuit is labelled with its name (S1-1, S1-2 …), phase (A, B or C), voltage, breaker, tile count and load; the *power source end* label goes on the distribution end and the *tile end* label on the first tile's **POWER IN**.
- Every label says **1/2** or **2/2**, DATA or POWER, and the cable name in very large type.

Options: **Cables to label** (data and power, data only or power only; shown only when both exist); **Label sheet** (Avery 5160, 30 per US Letter sheet; Avery 5163, 10 per US Letter sheet; Avery L7160, 21 per A4 sheet; a 62 × 29 mm label roll; or a custom roll size in mm, one label per page); **Copies of each end** (1 to 4); **Start at label no.** (skips labels already used on a part-used sheet); **Colour band** (a coloured edge per data port, or per power phase: A black, B red, C blue); **Also label the tile-to-tile links** (adds a label pair for every jumper between neighbouring tiles, such as P1-4 or S1-2-7); **Keep the two ends of a cable in one row** (sheets only: a cable's pair never splits across rows, so on a 3-across sheet the third cell of a row may stay empty).

Click **Print labels**. In the desktop app a preview page opens and the print window appears; choose your printer, set the scale to **100%** (turn off *fit to page*) and load the label sheet. In the mobile app **Save label sheet** saves the sheet as a web page through the share sheet; open it in a browser and print it at 100%. The line under the options shows how many cables, labels and pages will print. Labels print black on white with the colour bands; on a black-and-white printer the cable name still tells the ends apart.

19. Saving your work

Everything (canvas, walls, tile models, patterns, uploads, wiring, crew, options) is saved automatically and restored next time you open TileSmith. The desktop app keeps it in `%APPDATA%\TileSmith\state.json` (Windows) or `~/Library/Application Support/TileSmith/state.json` (Mac). To start fresh, open ⓘ **Info** and click **Reset saved settings**.

20. Keeping CPU use low

- Use **Fit canvas to walls** so NDI® does not encode empty canvas.
- Pick the lowest NDI® **Frame rate** your show needs; 30 needs roughly half the work of 60.
- Tick **Save CPU while the picture is still** if your receivers are happy holding the last frame.
- For an app on the same computer, Spout (Windows) or Syphon (Mac) costs far less CPU than NDI®, because the picture is not compressed. NDI®, Spout, Syphon and video export share one read of the canvas per frame, so running them together adds little.
- Animated patterns (sweep), Scroll, the build and the DVD screensaver redraw every frame. cycle redraws only when its colour changes and tileshade-cycle only while its colours slide; still pictures are redrawn only a few times a second.

21. Mobile app (Android and iOS)

The Android and iOS apps run the same TileSmith page as the desktop app, so canvas, walls, tile models, patterns, engineering, wiring, power and the build animation all work the same way, and everything is saved automatically on the device. They are built from the `mobile` folder (see **BUILD-ANDROID.txt** and **BUILD-IOS.txt**); a Kotlin version of the Android app for Google AI Studio is in the `android-aistudio` folder. Tablets give the most room; on a phone the panels stack under each other. The mobile app has its own manual, **TileSmith-Mobile-Manual.pdf**.

- **Saving files**: Export PNG, Export CSV, Spec sheet and Resolume XML open the share sheet, where you pick where the file goes: Files, Google Drive, iCloud, AirDrop, Quick Share, e-mail and so on. Closing the sheet cancels the save (TileSmith says *Not saved*).
- **Spec sheet** is saved as an HTML file; open it in a browser to print it or save it as a PDF.
- **Export video** records an H.264 MP4 with the device's own encoder, in real time: the Blank intro, the live build and the Final hold, the same as on the desktop. The screen stays on while it records; keep TileSmith open until it says *Saved*, then pick where the video goes ("Save Video" puts it in Photos on iPhone and iPad). HAP, ProRes and the DVD screensaver's **Export loop video** are desktop only.
- **NDI®, Spout and Syphon** are not in the mobile app yet, so their buttons and submenus are hidden, along with their notices in ⓘ Info.
- **Touch**: two fingers pinch to zoom and drag to pan the viewport. One finger drags a wall, wires tiles, marks half tiles and removes tiles, just like the mouse. Links (Ko-fi, ndi.video ...) open in the system browser.

22. Troubleshooting

Problem	What to do
"NDI® runtime not found"	TileSmith was built without the NDI® SDK. Install the NDI® SDK on the build computer and rebuild, or install the NDI® Runtime (ndi.link/NDIRedistV6, or NDIRedistV6Apple on Mac) or NDI® Tools on this computer, then open TileSmith again.

Problem	What to do
Receivers cannot see the NDI® source	Allow TileSmith through the firewall on private networks (Windows) or allow Local Network access (Mac). Make sure both computers are on the same network.
"No ffmpeg with the ... encoder found"	Windows: put the gyan.dev "full" ffmpeg.exe in the bin folder and rebuild. Mac: install ffmpeg with Homebrew (see BUILD-MAC.txt).
The app looks like an older version	Check the version in the window title. Rebuild with npm run clean, npm install, npm run dist, and run the new TileSmith-Portable.exe.
A wall will not move where I drag it	Walls snap to edges and each other. Hold Alt while dragging to place freely, or type the offset.
Data wiring shows ports over capacity	Check the processor, frame rate and bit depth, or use Reset to spec. In Manual wiring, start a New port.
Power options are missing	Tick Power distribution under Engineering › Power. While it is off, the power options and Power wiring are hidden.
"One tile draws more than a circuit can supply"	Check the Breaker (A) and Circuit voltage, and the tile model's Max watts.
NDI® frame rate is low	Read the timings in the NDI® status line. A large **read** or **send** time means the canvas is big: use Fit canvas to walls or a lower frame rate. With Save CPU on, a still picture is sent about 4 times a second on purpose. Plug a laptop into power and use a high-performance power mode.
No Start Spout / Start Syphon button	That copy was built without Spout or Syphon (the options submenu says so). Rebuild from a fresh copy of the project: bin\spout (Windows) and bin/mac/syphon (Mac) come in the zip, and the build prints [ok] Spout (or Syphon) output will be built into TileSmith. The web version never has the button.
The receiver lists TileSmith (Spout) but shows black	The receiver is on a different graphics card from the TileSmith sender (common on laptops with two graphics chips). Spout options › Graphics card: pick the card the receiver runs on (the status line shows the card TileSmith uses), or leave it on Automatic and set the receiver to High performance in Windows Settings › System › Display › Graphics.
"Spout could not send frames" or "could not start DirectX 11"	Pick another card under Spout options › Graphics card and update the graphics driver. Spout needs DirectX 11; it does not work over Remote Desktop or in most virtual machines. Details are in %APPDATA%\TileSmith\spout-log.txt.
"Syphon failed to load" or macOS says syphon.node or Syphon cannot be opened	The files were blocked by the download quarantine. Run xattr -cr /Applications/TileSmith.app in Terminal and open TileSmith again (the build clears it automatically).
A receiver does not list TileSmith	Spout and Syphon only reach apps on the same computer. Start it in TileSmith first, check the name in the options, and refresh the receiver's source list. Two senders cannot share a name.

Problem	What to do
Mobile: Export video says the device cannot record MP4	Update the system: on Android update "Android System WebView" and Chrome from Google Play (Android 7 cannot get a new enough version); on iPhone and iPad update iOS / iPadOS.
Mobile: a save does nothing	Pick an app in the share sheet (Files, Drive ...). Closing the sheet cancels the save.
Image looks soft or stretched	Change Image fit. Use Original size on canvas or on wall for pixel-exact images.
Settings are wrong after testing	ⓘ Info › Reset saved settings.

Build problems (npm warnings, vulnerabilities, PowerShell script errors, signing) are covered in the TROUBLESHOOTING sections of BUILD-WINDOWS.txt, BUILD-MAC.txt, BUILD-ANDROID.txt and BUILD-IOS.txt.

23. Version history

Version	Changes
3.3	Spout output (Windows) and Syphon output (Mac): share the master canvas with programs on the same computer through the graphics card. Start Spout / Start Syphon in the top bar (shown only when the app is built with them) and Spout options / Syphon options under Animation Options (name, frame rate when NDI® is off, Save CPU). SpoutDX and Syphon ship with the project and are built in automatically; credits and licences in ⓘ Info. NDI®, Spout and Syphon share one output timer and one canvas read per frame.
3.3	Wiring preferences: Port areas is now Data wiring preference and Circuit areas is now Wiring preference, each with two new choices, Horizontal and Vertical (bands of whole rows or whole columns across the wall, whichever way the pattern runs).
3.3	Android and iOS apps built from the same page (mobile folder, BUILD-ANDROID.txt and BUILD-IOS.txt): files are saved through the share sheet, Export video records H.264 MP4 with the device's encoder, NDI®, Spout and Syphon are hidden. Two-finger pinch zooms and pans the viewport (also on Windows touch screens).
3.2	Left panel reorganised: Canvas (LED walls, Tile model, Size of LED Wall, Wall position), Test pattern (Tile borders & overlays), Engineering (LED processor with Data wiring; Power with Power wiring) and Animation Options (DVD screensaver, Build options with Technician, Video export, NDI® options). Subtitle "LED Wall Pixel Map & Test Pattern Generator". Top bar: Next pattern, Start NDI® (replaces the NDI® pop-out window) and a Resolume XML drop-down.
3.2	Power planning: tick Power distribution (Socapex, 6 circuits per cable) for tiles per circuit from circuit voltage, breaker size and the 80% rule; Power wiring with Snake, Linear or Manual, circuit areas and a power link limit; circuits S1-1..., phases A/B/C, Socapex cables, spare circuits and load per phase in the engineering panel, spec sheet and CSV. Power runs are marked with a lightning bolt and circuit number; with data paths shown too, data runs left of the tile centres and power right, in white on a black casing.

Version	Changes
3.2	Manual data and power wiring by click or drag: fast drags fill every tile crossed and dragging back takes tiles off.
3.2	Wall shapes: half tiles for models with a half version (ROE Carbon CB3/CB5/CB8, YES TECH MU 500×1000, YES TECH MT II 1000×500): keep one half, add halves, join halves upright, offset by half a tile, or (ROE) on their side. Remove tiles for walls that are not rectangles. Tile position arrows (↑ row from the bottom, → column from the left) on any pattern.
3.2	Presets: 37 YES TECH models, ROE Carbon CB3 and the Carbon half panels; ROE, Absen and Unilumin presets corrected to the manufacturers' spec sheets (CB5 500 W, CB8 430 W and others); the preset list is grouped by manufacturer.
3.2	Patterns: tileshade-clean (no name, size or position boxes) and tileshade-cycle (red, green and blue sliding from colour to colour, never matching a neighbouring tile). The tileshade footer shows the wall name and position, and the position label sits on the first tile in a see-through box. Tile-shade color shows only for the tileshade patterns.
3.2	DVD screensaver (was DVD logo bounce) with a Perfect loop mode and Export loop video. Viewport: no labels around the canvas, zoom out far enough to see the whole canvas in any window size, and the top bar scrolls sideways in small windows. The fps counter only counts real animation.
3.1	Full speed when TileSmith is behind other windows or minimised (no Chromium backgrounding, no macOS App Nap while NDI® output or recording runs). Live frame timings in the NDI® status line and a canvas fps counter in the top bar.
3.0	NDI® runtime built into the desktop app from the NDI® SDK, with its license notices; no NDI® Tools or Runtime needed where TileSmith runs. Info dialog NDI® section with trademark statement, links and license note; NDI® marked with ® throughout. Fixed the universal Mac build merge step.
2.9	DVD logo bounce: the corner hit chance slider now tops out at about 45% of bounces.
2.8	DVD logo bounce with corner-hit chance. Video export formats: H.264 and H.265 (.mp4), ProRes 422 HQ and ProRes 4444 (.mov) alongside HAP. "Array" renamed "Size of LED Wall". This manual.
2.7	Removed the Barco and Analog Way processors and the scan, safe, crosshatch, checker and grid patterns. "bars" became SMPTEbars with Image fit. Built-in image patterns named by aspect ratio.
2.6	All audio removed. "Playback & audio" became "Build Animation Options".
2.5	Branded test patterns with text or a logo in the centre box.
2.4	Built-in disguise test patterns.
2.3	Scroll works on every pattern. "Original size" image fit modes. Uploads keep full resolution.
2.2	NovaLCT-style data wiring: 8 start-corner/direction patterns and port areas.
2.1	Much lower CPU use. NDI® frame rate and "Save CPU" options. NovaStar H series sending cards.
2.0	Multiple walls with drag and snap, LED processors, data wiring, power, engineering panel, Resolume export.